

GRADE 12 PHYSICS

EUEE MASTER STUDY GUIDE

Ethiopian University Entrance Exam

Complete Exam Preparation • All 5 Units
Mechanics • Fluids • Electromagnetism • Electronics
Simplified • Exam-Focused • High-Score Strategies

*Based on the Federal Democratic Republic of Ethiopia
Ministry of Education Grade 12 Physics Textbook (2023)*

HOW TO USE THIS GUIDE

This guide covers every topic in your Grade 12 Physics textbook, simplified and organised so you can study smarter — not harder. All formulas are written with the proper mathematical and physical symbols ($\times$, $\div$, $\sqrt{}$, Σ , $\int$, σ , ω , μ , Ω ...) so they look exactly as they would in an exam.

Here is what each box means:

EXAM FOCUS

Must-know content — these topics appear in almost every EUEE paper.

COMMON MISTAKES

The errors that cost students marks every single year. Read carefully — most students miss easy points here.

PRO TIP

Smart shortcuts, memory tricks, and time-savers used by top scorers.

QUICK REVISION

A 30-second summary of the section. Use it the night before the exam.

PRACTICE QUESTION

EUEE-style multiple-choice or short-answer question to test mastery.

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UNIT 1 — APPLICATION OF PHYSICS IN OTHER FIELDS

Physics is the foundation science. Almost every modern technology and other branch of science depends on physical laws. The EUEE often tests this unit through short conceptual questions, so you can score quick marks here.

1.1 Physics and Other Sciences

Key Definitions

- Physics — the science of matter, energy and their interactions.
- Spectroscopy — studying matter by analysing the light it emits or absorbs.
- Photon — a quantum (packet) of electromagnetic energy.
- Luminosity — the true brightness of a star (independent of distance).

Simplified Explanations

- Physics & Chemistry — atomic and subatomic physics explains chemical bonds, ionic/covalent interactions, periodicity and crystal structure.
- Physics & Biology — Newtonian mechanics explains body motion; fluid dynamics explains blood flow; sound waves explain hearing/speech; optics explains vision; electricity explains nerve signals.
- Physics & Astronomy — Newton's law of gravitation explains planetary motion (Kepler's laws); electromagnetic spectrum used by telescopes; atomic transitions explain spectral lines from stars.
- Physics & Geology — properties of rocks (density, magnetisation, elasticity, radioactivity) and seismic wave analysis depend on physics.

Inverse-square law of brightness: $\text{apparent brightness} \propto \frac{\text{luminosity}}{\text{distance}^2}$

EXAM FOCUS

- A light-year = distance light travels in 1 year $\approx 9.46 \times 10^{15}$ m.
- Photon emitted when electron jumps to a lower orbit; photon absorbed when it jumps to a higher orbit.
- Pascal's principle is used by soft-bodied animals (earthworms) to move.

1.2 Physics and Engineering

Key Points

- Civil Engineering — Newton's laws, statics & dynamics, materials science (used in bridges, buildings, dams).
- Mechanical Engineering — thermodynamics, fluid mechanics, mechanics (engines, turbines, machinery).
- Electrical & Electronic Engineering — electromagnetism, circuit theory, semiconductor physics.
- Aerospace Engineering — Bernoulli's principle, Newton's third law (lift and thrust).

PRO TIP

Remember the rule: STRUCTURE = STATICS, MOTION = DYNAMICS, FLOW = FLUIDS, POWER = ELECTROMAGNETISM.

1.3 Medical Physics

- X-rays — high-energy EM waves used to image bones (Wilhelm Röntgen, 1895).
- CT scan — combines many X-ray images for 3D images of organs.
- MRI — uses strong magnetic fields and radio waves; no ionising radiation.
- Ultrasound — high-frequency sound waves; safe for fetal imaging.
- Radiotherapy — gamma rays used to destroy cancer cells.
- Lasers — used in eye surgery (LASIK), tumour removal, dermatology.

COMMON MISTAKES

- Confusing CT scan (X-rays) with MRI (magnetic field). They use totally different physics.
- Forgetting that ultrasound is mechanical (sound) waves, not EM waves.

1.4 Physics and Defence Technology

- RADAR — radio detection and ranging using reflected microwaves.
- SONAR — sound navigation and ranging using ultrasonic waves underwater.
- GPS — satellites + relativity-corrected timing.
- Night vision — uses infrared radiation.
- Laser-guided weapons — based on light reflection and detection.

1.5 Physics in Communication

- Telephone, radio, TV, satellites, internet — all rely on electromagnetic waves.
- Optical fibres — total internal reflection of light carries digital signals.
- Mobile phones — modulated radio waves (microwave region).

QUICK REVISION — UNIT 1

Physics underpins chemistry, biology, astronomy, geology, all engineering, medicine, defence and communication.

Key tools: spectroscopy, X-rays, MRI, lasers, radar, sonar, GPS, optical fibres.

Inverse-square law: brightness $\propto 1/d^2$ — appears in light, sound, and gravity questions.

PRACTICE QUESTIONS — UNIT 1

1. Which medical imaging technique does NOT use ionising radiation? (a) X-ray (b) CT scan (c) MRI (d) PET scan
2. The physics principle behind RADAR is the reflection of _____ waves.
3. State two ways physics is used in geology.

Answers: 1) c 2) radio (microwave) 3) Seismic-wave analysis; measuring rock properties (density, magnetisation, radioactivity).

UNIT 2 — TWO-DIMENSIONAL MOTION

This is one of the highest-yield units on the EUEE. Expect questions on projectiles, circular motion, and gravitation almost every year.

2.1 Projectile Motion

Important Definitions

- Projectile — any object thrown into the air subject only to gravity (air resistance ignored).
- Trajectory — the curved path (a parabola) followed by a projectile.
- Range (R) — horizontal distance covered.
- Time of flight (T) — total time the projectile is in the air.
- Maximum height (H) — highest vertical position reached.

Key Equations (launch speed u , angle θ , $g = 9.8 \text{ m/s}^2$)

$$\text{Horizontal: } x = (u \cos\theta) t \qquad \text{Vertical: } y = (u \sin\theta) t - \frac{1}{2} g t^2$$

$$\text{Time of flight: } T = 2 u \sin\theta / g$$

$$\text{Maximum height: } H = u^2 \sin^2\theta / (2g)$$

$$\text{Range: } R = u^2 \sin(2\theta) / g$$

$$\text{Maximum range when } \theta = 45^\circ: R_{\text{max}} = u^2 / g$$

EXAM FOCUS

- Horizontal velocity is CONSTANT (no air resistance).
- Vertical velocity changes due to gravity at 9.8 m/s^2 downward.
- Two angles give the same range: θ and $(90^\circ - \theta)$. Example: 30° and 60° .
- Maximum range is at 45° .

COMMON MISTAKES

- Mixing horizontal and vertical components — they are independent!
- Forgetting the ' $\sin(2\theta)$ ' (not $\sin \theta$) in the range formula.
- Using $g = 10$ when the question says 9.8 (or vice versa).

PRO TIP

For a projectile launched and landing at the same height: 'time up = time down'. Use this to halve your work.

Worked Example

A ball is kicked at 20 m/s at 30° above the horizontal. Find R, H, T ($g = 10 \text{ m/s}^2$).

Solution:

$$T = 2(20)(\sin 30^\circ) / 10 = 2 \text{ s}$$

$$H = (20)^2 (\sin^2 30^\circ) / (2 \cdot 10) = 400 \cdot 0.25 / 20 = 5 \text{ m}$$

$$R = (20)^2 \cdot \sin 60^\circ / 10 = 400 \cdot 0.866 / 10 \approx 34.6 \text{ m}$$

2.2 Rotational Motion (Kinematics)

- Angular displacement θ (radians); 1 revolution = $2\pi \text{ rad} = 360^\circ$.
- Angular velocity $\omega = d\theta/dt$ (rad/s).
- Angular acceleration $\alpha = d\omega/dt$ (rad/s²).
- Linear-angular link: $v = r\omega$, $a_t = r\alpha$, $a_c = v^2/r = \omega^2 r$.

Equations of uniform angular acceleration

$$\omega = \omega_0 + \alpha t \qquad \theta = \omega_0 t + \frac{1}{2} \alpha t^2 \qquad \omega^2 = \omega_0^2 + 2\alpha\theta$$

2.3 Rotational Dynamics

- Moment of inertia I — rotational analogue of mass; depends on shape and axis.
- Torque $\tau = r F \sin\theta = I \alpha$ (Newton's second law for rotation).
- Rotational kinetic energy: $KE_{\text{rot}} = \frac{1}{2} I \omega^2$.
- Angular momentum: $L = I \omega$; conserved when net external torque = 0.

Common Moments of Inertia

Object	I about axis through centre
Solid disk / cylinder	$\frac{1}{2} M R^2$
Hollow cylinder (thin)	$M R^2$
Solid sphere	$(2/5) M R^2$
Thin rod (centre)	$(1/12) M L^2$

PRO TIP

A spinning skater pulling arms in spins faster — $I\omega$ is conserved (smaller I , larger ω).

2.4 Kepler's Laws of Planetary Motion

- 1st (Law of Orbits): Every planet moves in an ellipse with the Sun at one focus.
- 2nd (Law of Areas): Line from Sun to planet sweeps equal areas in equal times — planets move faster when nearer the Sun.
- 3rd (Law of Periods): $T^2 \propto r^3$ — square of the orbital period is proportional to cube of the average orbital radius.

$$T^2 / r^3 = 4\pi^2 / (G M) \quad (\text{constant for all planets around a given star})$$

2.5 Newton's Law of Universal Gravitation

$$F = G m_1 m_2 / r^2, \quad G = 6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$$

- Force is always attractive and acts along the line joining the two masses.
- Gravitational field strength: $g = G M / r^2$.
- Gravitational PE: $U = - G m_1 m_2 / r$.
- Escape velocity: $v_{\text{esc}} = \sqrt{2 G M / R}$.
- Orbital velocity: $v_{\text{orb}} = \sqrt{G M / R}$.
- Geostationary satellite altitude $\approx 36\,000 \text{ km}$ above the equator.

EXAM FOCUS

- $v_{\text{esc}} = \sqrt{2} \times v_{\text{orb}}$ (about $1.41 \times$ the orbital speed).
- On Earth's surface $g = 9.8 \text{ m/s}^2$; falls as $1/r^2$ with altitude.
- Weightlessness in orbit = continuous free-fall, NOT zero gravity.

QUICK REVISION — UNIT 2

Projectile: independent x and y motion; R_{max} at 45° .

Circular motion: $a_c = v^2/r$ toward the centre.

Rotational: $\tau = I\alpha$, $KE = \frac{1}{2}I\omega^2$, $L = I\omega$.

Kepler 3: $T^2 \propto r^3$. Newton: $F = Gm_1m_2/r^2$.

PRACTICE QUESTIONS — UNIT 2

1. A projectile is fired at 50 m/s at 53° . Find the maximum height ($g=10$).
 2. A wheel accelerates from rest to 30 rad/s in 6 s . Find its angular acceleration.
 3. The Earth's orbit radius is 1 AU and period 1 year . Mars is 1.52 AU . Find Mars' year.
- Answers: 1) $H = (50^2 \cdot \sin^2 53^\circ) / (20) \approx 80 \text{ m}$ 2) $\alpha = 5 \text{ rad/s}^2$ 3) $T = \sqrt{(1.52^3)} \approx 1.87 \text{ yr}$.

UNIT 3 — FLUID MECHANICS

3.1 Fluid Statics

- Fluid — substance that flows; includes liquids and gases.
- Density: $\rho = m / V$ (kg/m^3). Water = 1000 kg/m^3 .
- Specific gravity (relative density): $\rho_{\text{substance}} / \rho_{\text{water}}$ — has no units.
- Cohesion — attraction between like molecules (water-water).
- Adhesion — attraction between unlike molecules (water-glass).
- Surface tension — the energy stored per unit area at a fluid surface.
- Capillarity — rise/fall of liquid in narrow tubes due to surface tension.
- Viscosity (η) — resistance of a fluid to flow.

3.2 Pressure in Fluids at Rest

$$\text{Pressure: } P = F / A \quad (\text{Pa} = \text{N/m}^2)$$

$$\text{Hydrostatic pressure at depth } h: P = \rho g h$$

$$\text{Absolute pressure: } P_{\text{abs}} = P_{\text{atm}} + \rho g h$$

Pascal's principle: pressure applied to a confined fluid is transmitted equally in all directions.

$$\text{Hydraulic press: } F_1 / A_1 = F_2 / A_2$$

EXAM FOCUS

- $1 \text{ atm} = 1.013 \times 10^5 \text{ Pa} = 760 \text{ mmHg}$.
- Hydrostatic pressure depends on depth and density only — not container shape.
- Hydraulic systems multiply force, not energy ($W_1 = W_2$).

3.3 Archimedes' Principle

Any object wholly or partly immersed in a fluid is buoyed up by a force equal to the weight of the fluid it displaces.

$$\text{Buoyant force: } F_b = \rho_{\text{fluid}} \cdot V_{\text{displaced}} \cdot g$$

- If $F_b > \text{weight}$ → object floats (rises).
- If $F_b = \text{weight}$ → object suspends (neutral buoyancy).
- If $F_b < \text{weight}$ → object sinks.
- Apparent weight in fluid = real weight — buoyant force.

COMMON MISTAKES

- Using object's volume instead of displaced volume when the object is partly submerged.
- Forgetting that the buoyant force equals weight of displaced FLUID (not the object).

3.4 Fluid Flow

- Streamline (laminar) flow — smooth layers, no mixing.
- Turbulent flow — irregular, eddies; high Reynolds number.
- Equation of continuity: $A_1 v_1 = A_2 v_2$ (mass-conserved, incompressible).
- Bernoulli's equation: $P + \frac{1}{2}\rho v^2 + \rho g h = \text{constant along a streamline.}$

$$P_1 + \frac{1}{2} \rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2} \rho v_2^2 + \rho g h_2$$

PRO TIP

Bernoulli in one sentence: 'fast fluid = low pressure'. Aeroplane lift, perfume sprays, and curving footballs all rely on this!

3.5 Safety and High Pressure

- Pressure cookers cook faster (higher boiling point at higher pressure).
- Deep-sea divers must rise slowly to avoid the bends (nitrogen bubbles in blood).
- Aircraft cabins are pressurised because external pressure is too low at altitude.
- Hydraulic brakes use Pascal's principle.

QUICK REVISION — UNIT 3

$P = F/A$, $P = \rho g h$, $F_b = \rho V g$, $A_1 v_1 = A_2 v_2$, Bernoulli: $P + \frac{1}{2}\rho v^2 + \rho g h = \text{const.}$
Pascal's principle → hydraulics. Archimedes → buoyancy. Bernoulli → flow.

PRACTICE QUESTIONS — UNIT 3

1. A diver is 20 m below the surface ($\rho=1000$, $g=10$, $P_{\text{atm}}=10^5$ Pa). Find absolute pressure.
2. A 200 cm^3 object is fully submerged in water. Find buoyant force.
3. Pipe narrows from 8 cm^2 to 2 cm^2 . If $v_1 = 1 \text{ m/s}$, find v_2 .

Answers: 1) $P = 10^5 + 1000 \cdot 10 \cdot 20 = 3 \times 10^5 \text{ Pa}$ 2) $F_b = 1000 \cdot 2 \times 10^{-4} \cdot 10 = 2 \text{ N}$ 3) $v_2 = 4 \text{ m/s}$.

UNIT 4 — ELECTROMAGNETISM

4.1 Magnets and Magnetic Field

- Magnet — material that attracts iron, nickel, cobalt.
- Every magnet has two poles: North (N) and South (S).
- Like poles repel; unlike poles attract.
- Magnetic field B (tesla, T) — region around a magnet where magnetic force acts.
- $1 \text{ T} = 10^4 \text{ gauss}$.

4.2 Magnetic Field Lines

- Lines run from N to S outside the magnet, S to N inside.
- Field lines never cross.
- Closer lines = stronger field.
- Earth itself is a giant magnet (geographic N pole = magnetic S).

4.3 Current and Magnetism

A moving charge or current produces a magnetic field (Oersted's experiment).

$$\text{Long straight wire: } B = \mu_0 I / (2\pi r)$$

$$\text{Centre of circular loop (N turns): } B = \mu_0 N I / (2 R)$$

$$\text{Inside a solenoid (n turns/m): } B = \mu_0 n I$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$$

Force on a current-carrying conductor in a magnetic field:

$$F = B I L \sin\theta$$

Force on a moving charge:

$$F = q v B \sin\theta \quad (\text{Lorentz force})$$

PRO TIP — RIGHT-HAND RULES

- Wire field: thumb = current, fingers curl = field direction.
- Force on current/charge: flat right hand — thumb (force), index (B-field), middle (current/v).
- For NEGATIVE charge, reverse the direction.

4.4 Electromagnetic Induction

- Induction — generating an EMF by changing magnetic flux through a coil.
- Magnetic flux: $\Phi = B A \cos\theta$ (weber, Wb).
- Faraday discovered induction in 1831.

4.5 Faraday's & Lenz's Laws

$$\text{Faraday's law: } \varepsilon = - N \cdot d\Phi/dt$$

Lenz's law: the induced current flows in the direction that opposes the change producing it (conservation of energy).

EXAM FOCUS

- EMF is induced ONLY when flux is CHANGING.
- A bar magnet pushed into a coil → current flows; held still → no current.
- Lenz's law gives the negative sign in Faraday's equation.

4.6 Transformers

- Device that transfers electrical energy between circuits via electromagnetic induction.
- Step-up: more turns on secondary → higher voltage.
- Step-down: fewer turns on secondary → lower voltage.

$$V_s / V_p = N_s / N_p \quad (\text{Ideal}) \quad V_p I_p = V_s I_s$$

COMMON MISTAKES

- Saying transformers work on DC — they need CHANGING flux, so AC only.
- Forgetting that step-up voltage means step-DOWN current ($P = VI$ conserved).

4.7 Application and Safety

- Generators — convert mechanical energy to electrical (induction).
- Motors — convert electrical to mechanical (force on current).
- Power transmission uses very high voltage to reduce I^2R losses.
- Earthing, fuses and circuit breakers prevent electric shock and fires.

QUICK REVISION — UNIT 4

B around wire: $\mu_0 I / (2\pi r)$. Force: $F = BIL \sin\theta$, $F = qvB \sin\theta$.

Faraday: $\varepsilon = -N d\Phi/dt$. Lenz: induced current opposes change.

Transformer: $V_s/V_p = N_s/N_p$ (AC only).

PRACTICE QUESTIONS — UNIT 4

1. A 0.5 m wire carries 4 A perpendicular to a 0.2 T field. Find force on it.
 2. A coil of 200 turns has flux change 0.01 Wb in 0.5 s. Find induced EMF.
 3. A transformer has $N_p = 100$, $N_s = 500$, $V_p = 220$ V. Find V_s .
- Answers: 1) $F = BIL = 0.2 \cdot 4 \cdot 0.5 = 0.4$ N 2) $\varepsilon = 200 \cdot 0.01 / 0.5 = 4$ V 3) $V_s = 220 \cdot 5 = 1100$ V.

UNIT 5 — BASIC ELECTRONICS

5.1 Semiconductors

- Conductor — many free electrons (Cu, Al). Insulator — almost none (rubber, glass).
- Semiconductor — between the two; e.g. silicon (Si), germanium (Ge).
- Intrinsic semiconductor — pure Si or Ge.
- Extrinsic semiconductor — doped to increase conductivity.
- n-type — doped with pentavalent atoms (P, As); free electrons are majority carriers.
- p-type — doped with trivalent atoms (B, Al, Ga); 'holes' are majority carriers.

5.2 Diodes and Their Functions

- Diode — formed by joining p-type and n-type semiconductors (p-n junction).
- Forward bias — p to (+), n to (–); current flows easily.
- Reverse bias — p to (–), n to (+); almost no current (until breakdown).
- Special diodes: LED (light), Zener (voltage regulation), Photodiode (light → current).

PRO TIP

Diode = one-way valve for current. Conventional current flows from p (anode) to n (cathode) when forward-biased.

5.3 Rectification

- Converting AC to DC using diodes.
- Half-wave rectifier — uses 1 diode; only half the AC cycle passes.
- Full-wave rectifier — uses 2 diodes (centre-tap) or 4 diodes (bridge).
- Bridge rectifier — most common; uses 4 diodes, no centre-tap transformer.
- Smoothing capacitors flatten the pulsating DC.

5.4 Transistors and Their Application

- Transistor — three-layer semiconductor device: emitter (E), base (B), collector (C).
- Two main types: NPN and PNP.
- Acts as a switch (digital) or amplifier (analogue).
- Current relation: $I_E = I_B + I_C$ (with I_B very small).
- Current gain (β) = I_C / I_B (typically 50–300).

5.5 Integrated Circuits (ICs)

- Many transistors, resistors, capacitors fabricated on a single silicon chip.
- Examples: 555 timer, 741 op-amp, microprocessors.
- Advantages: small size, low cost, high reliability, low power use.

5.6 Logic Gates and Logic Circuits

Gate	Symbol Equation	Output Y
AND	$Y = A \cdot B$	1 only if A=1 AND B=1

OR	$Y = A + B$	1 if A=1 OR B=1
NOT	$Y = \bar{A}$	Inverts input
NAND	$Y = (A \cdot B)'$	Inverse of AND
NOR	$Y = (A + B)'$	Inverse of OR
XOR	$Y = A \oplus B$	1 if $A \neq B$
XNOR	$Y = (A \oplus B)'$	1 if $A = B$

EXAM FOCUS

- NAND and NOR are 'universal gates' — every other logic function can be built from them.
- Boolean algebra: $A + 1 = 1$, $A \cdot 0 = 0$, $A + \bar{A} = 1$, $A \cdot \bar{A} = 0$.

5.7 Application of Electronics

- Computers, mobile phones, calculators, TVs.
- Industrial automation and control systems.
- Medical devices: ECG, pacemakers, MRI controllers.
- Renewable energy: solar inverters, smart grids.

QUICK REVISION — UNIT 5

n-type → electrons, p-type → holes. Diode = one-way valve. Rectifier: AC→DC.
Transistor: $I_E = I_B + I_C$; $\beta = I_C/I_B$. NAND/NOR = universal gates.

PRACTICE QUESTIONS — UNIT 5

1. Doping silicon with phosphorus produces a _____-type semiconductor.
 2. In a bridge rectifier, how many diodes are used?
 3. A transistor has $I_B = 50 \mu\text{A}$ and $\beta = 100$. Find I_C .
- Answers: 1) n 2) 4 3) $I_C = 100 \cdot 50 \mu\text{A} = 5 \text{ mA}$.

LAST-MINUTE REVISION GUIDE

Read this in the final 24 hours before the EUEE.

Mechanics

- $v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$.
- Projectile: $T = 2u \sin\theta/g$, $R = u^2 \sin 2\theta/g$, $H = u^2 \sin^2\theta/(2g)$.
- Circular: $a_c = v^2/r$, $F_c = mv^2/r$.
- Rotational: $\tau = I\alpha$, $KE = \frac{1}{2}I\omega^2$, $L = I\omega$.
- Gravitation: $F = Gm_1m_2/r^2$; $g = GM/r^2$; $v_{esc} = \sqrt{(2GM/R)}$.

Fluids

- $\rho = m/V$, $P = F/A$, $P = \rho gh$.
- Pascal: $F_1/A_1 = F_2/A_2$.
- Archimedes: $F_b = \rho Vg$.
- Continuity: $A_1v_1 = A_2v_2$.
- Bernoulli: $P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$.

Electromagnetism

- B around wire $= \mu_0 I/(2\pi r)$; inside solenoid $B = \mu_0 nI$.
- $F = BIL \sin\theta$; $F = qvB \sin\theta$.
- $\Phi = BA \cos\theta$; Faraday $\varepsilon = -N d\Phi/dt$.
- Transformer: $V_s/V_p = N_s/N_p$; $V_p I_p = V_s I_s$.

Electronics

- n-type = electrons, p-type = holes.
- Diode forward bias $\rightarrow$ conducts; reverse bias $\rightarrow$ blocks.
- Bridge rectifier uses 4 diodes.
- Transistor: $I_E = I_B + I_C$; $\beta = I_C/I_B$.
- NAND and NOR are universal gates.

EUEE EXAM STRATEGY

Time Management

- Total time: divide it by number of questions to set a target per question.
- First pass: answer all easy questions you are sure about (~60% of time).
- Second pass: tackle medium questions (~30%).
- Third pass: attempt hard questions and review answers (~10%).
- Never spend more than 2 minutes on a single question on first pass.

Answering Techniques

- Read every option before choosing — don't pick the first 'reasonable' one.
- Eliminate clearly wrong options first; this raises the odds of guessing correctly.
- Watch the units — many questions trick you with wrong units (cm vs m, min vs s).
- Sketch a quick diagram for every mechanics, optics or electromagnetism problem.
- Check arithmetic by quick estimation (orders of magnitude).

Common Traps

- Sign of acceleration in vertical motion (negative when up is positive).
- Mass vs weight (weight = mg , in newtons).
- Series vs parallel resistor combinations.
- Real image vs virtual image in optics (which side, inverted or not).
- Forgetting $\cos \theta$ or $\sin \theta$ in flux/work problems.
- 'Per unit length' vs 'total' in solenoid problems.

PRO TIP

If two options are mathematical opposites (+ and –) one of them is usually correct. If two options are nearly identical with one tiny change, look very carefully — examiners use this to test understanding.

MOST IMPORTANT TOPICS TO REVISE

Based on past EUEE patterns, focus on these high-yield topics:

Rank	Topic	Why it matters
★★★★★	Projectile motion	Almost every year; numerical.
★★★★★	Newton's gravitation & Kepler's laws	Conceptual + numerical.
★★★★★	Pascal's principle & Archimedes	Quick numerical wins.
★★★★★	Faraday's & Lenz's laws	Frequently tested concept.
★★★★★	Transformers	Easy formula application.
★★★★	Rotational dynamics (torque, I , ω)	Conceptual.
★★★★	Bernoulli's equation	Plane lift, sprayer questions.
★★★★	Force on current/charge in B-field	$F = BIL$, $F = qvB$.
★★★★	Diodes & rectifiers	Identify circuits.
★★★	Logic gates (truth tables)	Easy if memorised.

NEXT-LEVEL LEARNING (BONUS)

These are slightly advanced ideas from first-year university physics. They are NOT required for the EUEE, but knowing them gives you confidence and deeper understanding — and may help with tricky conceptual questions.

1. Special Relativity (Einstein, 1905)

Time dilation: $\Delta t = \Delta t_0 / \sqrt{1 - v^2/c^2}$. Mass-energy equivalence: $E = mc^2$. As speed approaches the speed of light, time slows down and mass increases.

2. Quantum Mechanics

Light behaves both as a wave and a particle (photon). Electrons have wavelike properties (de Broglie wavelength $\lambda = h/p$). The Heisenberg uncertainty principle: $\Delta x \cdot \Delta p \geq \hbar/2$.

3. Maxwell's Equations

Four equations describing all classical electromagnetism. They predict electromagnetic waves travel at $c = 1/\sqrt{\mu_0\epsilon_0}$. Light is an electromagnetic wave!

4. Thermodynamics: Entropy

The 2nd law: total entropy of an isolated system always increases. This explains why heat flows from hot to cold, and why time has a direction.

5. Modern Astronomy

Black holes (escape velocity $> c$), the Big Bang (universe expanding from a hot dense state ≈ 13.8 billion years ago), and dark matter (invisible mass that gravitationally affects galaxies).

PRO TIP

Linking what you study now to these advanced ideas builds intuition that wins conceptual EUEE questions and impresses examiners in interviews.

MASTER FORMULA SHEET

Mechanics (1-D & 2-D)

$$v = u + at, \quad s = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2as$$

$$\text{Projectile: } T = 2u \sin\theta/g, \quad H = u^2 \sin^2\theta/(2g), \quad R = u^2 \sin 2\theta/g$$

$$\text{Centripetal: } a_c = v^2/r = \omega^2 r, \quad F_c = mv^2/r$$

$$\text{Angular: } \omega = \omega_0 + \alpha t, \quad \theta = \omega_0 t + \frac{1}{2}\alpha t^2, \quad \omega^2 = \omega_0^2 + 2\alpha\theta$$

$$\text{Rotational dynamics: } \tau = I\alpha, \quad L = I\omega, \quad KE_{\text{rot}} = \frac{1}{2}I\omega^2$$

$$\text{Gravitation: } F = Gm_1m_2/r^2, \quad g = GM/r^2, \quad U = -Gm_1m_2/r$$

$$v_{\text{orb}} = \sqrt{GM/R}, \quad v_{\text{esc}} = \sqrt{2GM/R}$$

$$\text{Kepler 3: } T^2 \propto r^3$$

Fluids

$$\rho = m/V, \quad P = F/A, \quad P = \rho gh, \quad P_{\text{abs}} = P_{\text{atm}} + \rho gh$$

$$\text{Pascal: } F_1/A_1 = F_2/A_2$$

$$\text{Archimedes: } F_b = \rho_{\text{fluid}} V_{\text{disp}} g$$

$$\text{Continuity: } A_1 v_1 = A_2 v_2$$

$$\text{Bernoulli: } P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$$

Electromagnetism

$$B (\text{wire}) = \mu_0 I / (2\pi r), \quad B (\text{solenoid}) = \mu_0 n I, \quad B (\text{loop}) = \mu_0 N I / (2R)$$

$$\text{Force: } F = BIL \sin\theta, \quad F = qvB \sin\theta$$

$$\text{Flux: } \Phi = BA \cos\theta$$

$$\text{Faraday: } \varepsilon = -N d\Phi/dt$$

$$\text{Transformer: } V_s/V_p = N_s/N_p, \quad V_{\text{pl}}/I_p = V_{\text{sl}}/I_s$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$$

Electronics

$$\text{Ohm: } V = IR; \quad \text{Power: } P = VI = I^2R = V^2/R$$

$$\text{Transistor: } I_E = I_B + I_C, \quad \beta = I_C/I_B$$

$$\text{Boolean: } A + 1 = 1, \quad A \cdot 0 = 0, \quad A + \bar{A} = 1, \quad A \cdot \bar{A} = 0$$

Useful Constants

- $g = 9.8 \text{ m/s}^2$ (use 10 m/s^2 if asked)
- $G = 6.674 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$
- $c = 3 \times 10^8 \text{ m/s}$
- $e = 1.6 \times 10^{-19} \text{ C}$
- $k_e = 9 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$
- $\mu_0 = 4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$

- $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$
- $h = 6.63 \times 10^{-34} \text{ J}\cdot\text{s}$
- $1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$
- $\rho_{\text{water}} = 1000 \text{ kg/m}^3$

FINAL WORDS — YOU'VE GOT THIS!

You now hold a complete, exam-focused tool kit for the Grade 12 Physics EUEE. Read each unit slowly, then revise using the Quick Revision boxes and Master Formula Sheet. Practise the example questions until you can solve them without looking.

Remember: the EUEE rewards CLARITY and SPEED. Master the formulas, recognise the question type, and trust your preparation.

Good luck — score high, dream higher!